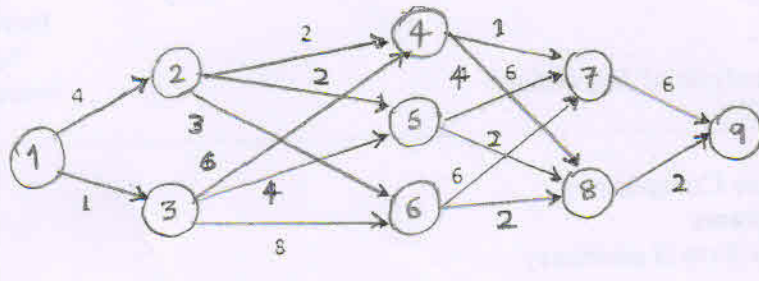




|                                                                                                                                                                                                                                                                       |                                                                                                                          |      |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------|----|----|----|----|---|---|---|----|---|----|---|----|---|---|---|---|
| Q 3. Solve any two.                                                                                                                                                                                                                                                   |                                                                                                                          |      |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q3 (a)                                                                                                                                                                                                                                                                | Find a minimum cost path from 1 to 9 in the given graph using dynamic programming using forward approach.                | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
|                                                                                                                                                                                     |                                                                                                                          |      |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q3 (b)                                                                                                                                                                                                                                                                | Explain Flow-shop scheduling with an example.                                                                            | 10M  |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q3 (c)                                                                                                                                                                                                                                                                | Find the path of Travelling Sales-person of given graph. <i>using dynamic programming</i>                                | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| <table border="1" data-bbox="672 792 891 938"> <tr><td>0</td><td>10</td><td>18</td><td>20</td></tr> <tr><td>5</td><td>0</td><td>9</td><td>10</td></tr> <tr><td>6</td><td>13</td><td>0</td><td>12</td></tr> <tr><td>8</td><td>8</td><td>9</td><td>0</td></tr> </table> |                                                                                                                          |      | 0  | 10 | 18 | 20 | 5 | 0 | 9 | 10 | 6 | 13 | 0 | 12 | 8 | 8 | 9 | 0 |
| 0                                                                                                                                                                                                                                                                     | 10                                                                                                                       | 18   | 20 |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| 5                                                                                                                                                                                                                                                                     | 0                                                                                                                        | 9    | 10 |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| 6                                                                                                                                                                                                                                                                     | 13                                                                                                                       | 0    | 12 |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| 8                                                                                                                                                                                                                                                                     | 8                                                                                                                        | 9    | 0  |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q 4. Solve any two.                                                                                                                                                                                                                                                   |                                                                                                                          |      |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q4 (a)                                                                                                                                                                                                                                                                | Solve the following sum of subset problem & Draw portion of state space tree for M=35.<br>W = { 5,7,10,12,15,18,20}      | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q4 (b)                                                                                                                                                                                                                                                                | Find out the longest common sub-sequence from the given two sequences:<br>P= (100101101)<br>Q= (01110)                   | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q4 (c)                                                                                                                                                                                                                                                                | With suitable example explain difference in pattern matching techniques of Brute-force and Knuth-Morris-Pratt Algorithm. | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q 5. Write a short note on any two.                                                                                                                                                                                                                                   |                                                                                                                          |      |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q5 (a)                                                                                                                                                                                                                                                                | 15 Puzzle Problem                                                                                                        | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q5 (b)                                                                                                                                                                                                                                                                | 8-Queen Problem                                                                                                          | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q5 (c)                                                                                                                                                                                                                                                                | Randomized Algorithm                                                                                                     | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |
| Q5 (d)                                                                                                                                                                                                                                                                | Bellman Ford Algorithm                                                                                                   | 10 M |    |    |    |    |   |   |   |    |   |    |   |    |   |   |   |   |